

(a) Shared excerpts



Inverse queries exchange the correct answer

$q_+ : \text{dog followed by rooster} \rightarrow W_+$

$q_- : \text{rooster followed by dog} \rightarrow W_-$

(b) ReTwin: two-stage fine-tuning

1 Query-conditioned competition

Score complete timestamp answers

	$y(W_+)$	$y(W_-)$
q_+	$s_{++} \uparrow$	$s_{+-} \downarrow$
q_-	$s_{-+} \downarrow$	$s_{--} \uparrow$

Candidate CE + label-exchange JS

2 Set-level preference learning

Compare coverage, count, and boundaries



Quality scores $\rightarrow$ soft preference targets

TRAINING

Inference

Audio + query $\rightarrow$ Adapted backbone $\rightarrow$ Timestamps

(c) Query-specific selection

Inverse-query grounding example

Paired SFT

+ Candidate

